Overview

Milestone 3 of the **TraceFinder** project focuses on developing a **deep learning–based scanner identification system** using a **Convolutional Neural Network (CNN)**.
The system is designed to learn **scanner-specific intrinsic artifacts**—such as noise patterns, texture inconsistencies, and sensor characteristics—from scanned document images.

To ensure **model transparency and forensic reliability**, **Grad-CAM (Gradient-weighted Class Activation Mapping)** is applied to visually explain **which regions of the document images influence the CNN's predictions**.

This milestone demonstrates the **end-to-end pipeline** of:

- CNN model training
- Performance evaluation using a confusion matrix
- Model explainability through visual heatmaps

Objectives

- Train a **CNN classifier** to identify the **source scanner** of a document image
- Achieve **training accuracy ≥ 85%** despite limited data
- Analyze classification performance using a **confusion matrix**
- Apply **Grad-CAM** to verify that the model focuses on **scanner artifacts rather than document content**

Dataset Description

- All images are stored directly in the **wiki/** directory
- **Class labels are automatically extracted from filenames**
- Naming convention:

scannerA_doc1.jpg → scannerA

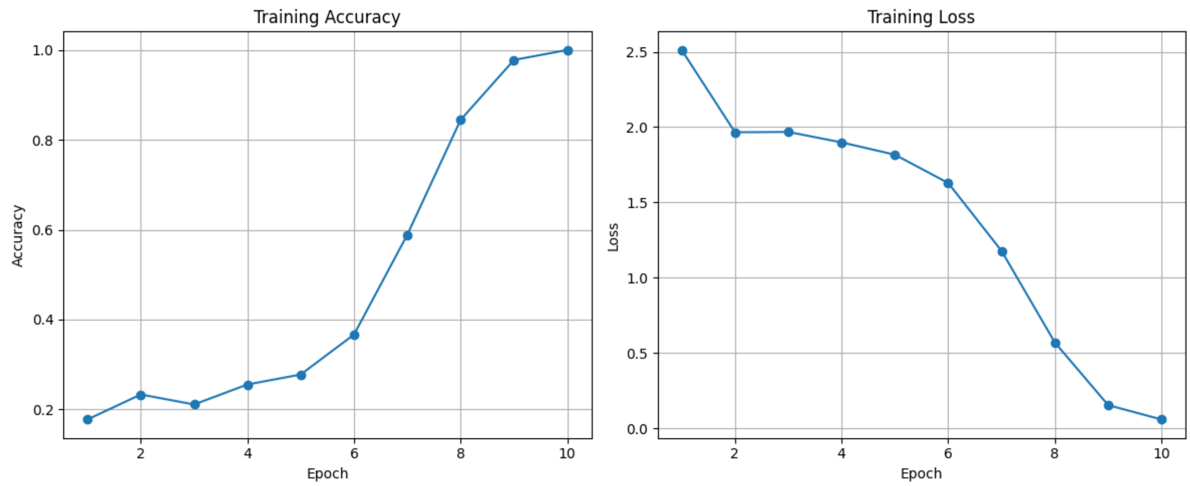
Train the CNN Model

✓ Outputs

- Training accuracy printed in console
- Trained model saved as:

models/cnn_scanner.h5

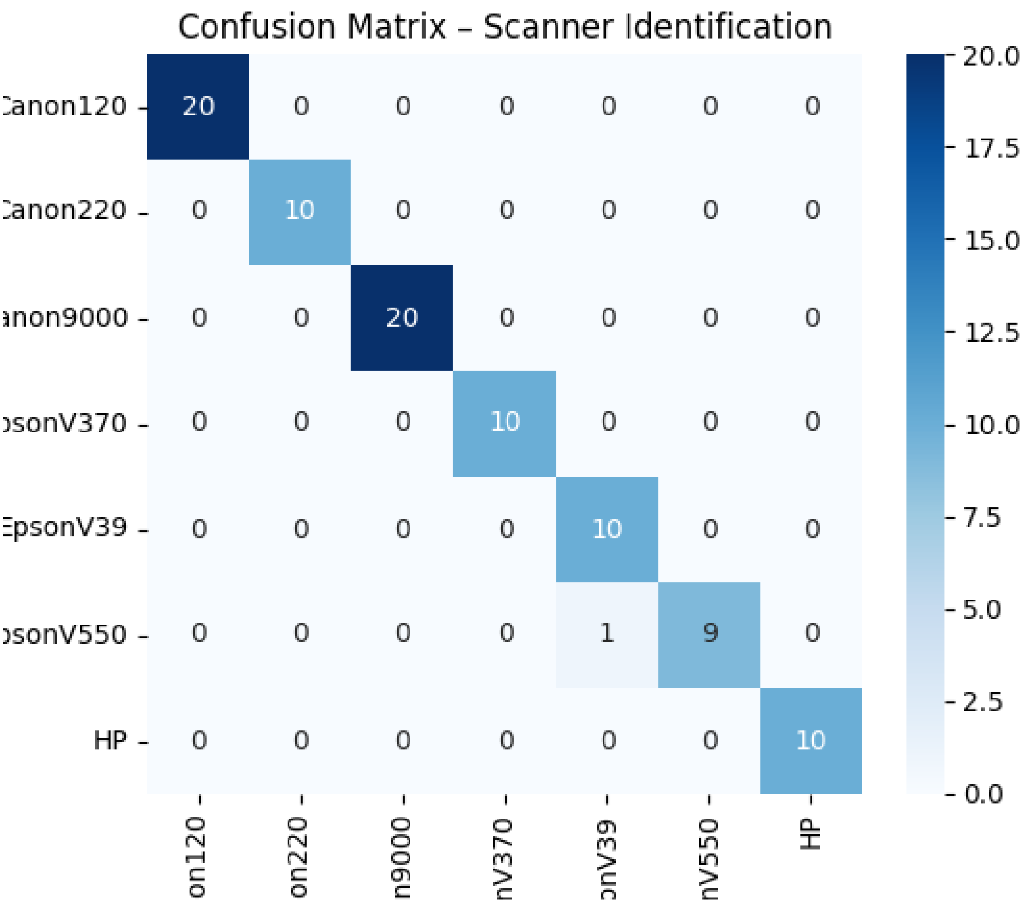
Train accuracy Curve



Generate Confusion Matrix

Outputs

results/confusion_matrix.png



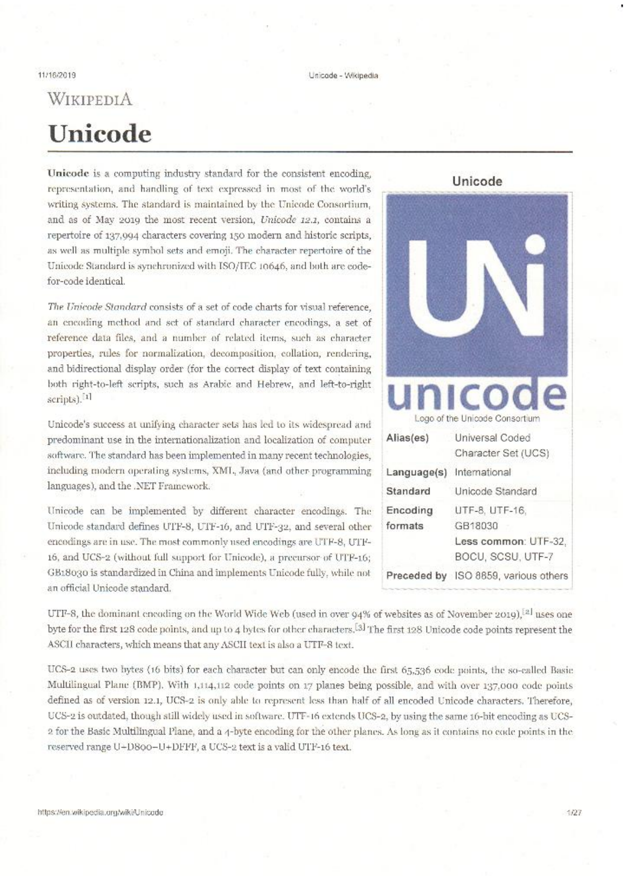
The confusion matrix provides:

- Per-scanner classification performance

- Error distribution across scanner classes

Grad-CAM Explainability

Sample



11/16/2019 Unicode - Wikipedia

WIKIPEDIA

Unicode

Unicode is a computing industry standard for the consistent encoding, representation, and handling of text expressed in most of the world's writing systems. The standard is maintained by the Unicode Consortium, and as of May 2019 the most recent version, *Unicode 12.1*, contains a repertoire of 137,994 characters covering 150 modern and historic scripts, as well as multiple symbol sets and emoji. The character repertoire of the Unicode Standard is synchronized with ISO/IEC 10646, and both are code-for-code identical.

The Unicode Standard consists of a set of code charts for visual reference, an encoding method and set of standard character encodings, a set of reference data files, and a number of related items, such as character properties, rules for normalization, decomposition, collation, rendering, and bidirectional display order (for the correct display of text containing both right-to-left scripts, such as Arabic and Hebrew, and left-to-right scripts).[1]

Unicode's success at unifying character sets has led to its widespread and predominant use in the internationalization and localization of computer software. The standard has been implemented in many recent technologies, including modern operating systems, XML, Java (and other programming languages), and the .NET Framework.

Unicode can be implemented by different character encodings. The Unicode standard defines UTF-8, UTF-16, and UTF-32, and several other encodings are in use. The most commonly used encodings are UTF-8, UTF-16, and UCS-2 (without full support for Unicode), a precursor of UTF-16; GB18030 is standardized in China and implements Unicode fully, while not an official Unicode standard.

Unicode

Logo of the Unicode Consortium

Alias(es)	Universal Coded Character Set (UCS)
Language(s)	International
Standard	Unicode Standard
Encoding formats	UTF-8, UTF-16, GB18030 **Less common**: UTF-32, BOCU, SCSU, UTF-7
Preceded by	ISO 8859, various others

UTF-8, the dominant encoding on the World Wide Web (used in over 94% of websites as of November 2019),[2] uses one byte for the first 128 code points, and up to 4 bytes for other characters.[3] The first 128 Unicode code points represent the ASCII characters, which means that any ASCII text is also a UTF-8 text.

UCS-2 uses two bytes (16 bits) for each character but can only encode the first 65,536 code points, the so-called Basic Multilingual Plane (BMP). With 1,114,112 code points on 17 planes being possible, and with over 137,000 code points defined as of version 12.1, UCS-2 is only able to represent less than half of all encoded Unicode characters. Therefore, UCS-2 is outdated, though still widely used in software. UTF-16 extends UCS-2, by using the same 16-bit encoding as UCS-2 for the Basic Multilingual Plane, and a 4-byte encoding for the other planes. As long as it contains no code points in the reserved range U+D800–U+DFFF, a UCS-2 text is a valid UTF-16 text.

https://en.wikipedia.org/wiki/Unicode 1/27

Outputs

Gradcam_output

Milestone 3 successfully demonstrates:

- Effective **CNN-based scanner identification**

- Strong performance despite limited training data
- Reliable **model explainability** using Grad-CAM

This milestone establishes a **robust foundation for forensic document analysis**, and sets the stage for:

- Larger datasets
- Cross-scanner generalization
- Advanced explainability and attribution methods